



GE Medical Systems

Technical Publications

GEMS Part Number 5113474
Revision A

TEAL PDU Module in HFD Cabinet

Copyright© 2001 by General Electric Company

Operating Documentation

TABLE OF CONTENTS

IMPORTANT SAFETY INSTRUCTIONS	3
1- INTRODUCTION	4
1-1 General Description.....	4
1-2 Scope	4
2- SPECIFICATIONS	4
2-1 Electrical Specifications.....	5
2-2 Environmental Specifications	5
2-3 Mechanical Specifications	5
3- INSTALLATION AND OPERATION.....	6
3-1 Installation	6
3-1.1 Input Voltage Selection	6
3-1.2 Connection	7
3-2 Operation.....	8
3-2.1 Main Input Circuit Breaker	8
3-2.2 Power Off	9
3-2.3 EMO Reset.....	9
3-2.4 TVSS Indicators	9
3-2.5 Phase Indicators	9
3-2.6 Phase Reversal Relay	9
3-2.7 J5, J7, & J8 Pass-thru Connector	9
3-2.8 J6 Pass-thru Connector	9
3-2.9 Input / Output Voltage Test Points	10
4- FIELD REPLACEABLE UNITS (FRU'S).....	12
4-1 FRU Installation	12
4-1.1 Replacement of the TVSS Module.....	12
4-1.2 Replacement of the Control PC Board	13
4-1.2 Replacement of the SGA PC Board.....	13
4-1.3 Troubleshooting Guide.....	14
SCHEMATIC DIAGRAMS.....	16-17
REVISION HISTORY.....	18

IMPORTANT SAFETY INSTRUCTIONS

SAVE THESE INSTRUCTIONS – This manual contains important instruction for the ACGD Power Distribution Unit (PDU) that must be followed during installation, operation, and maintenance.



⚠ DANGER



Electrical shock hazard. Opening enclosure exposes hazardous voltages. Refer service to qualified personnel.

⚠ IMPORTANT

As standards, specifications, and designs change from time to time, please ask for confirmation of the information given in this publication.

1- INTRODUCTION

1-1 General Description

The HFD Power Distribution Unit (PDU) distributes power to various components of a General Electric Medical Systems MRI system while providing surge suppression and electrical isolation for the power line. Three phase input voltage is selectable among six levels: 200/208/380/400/415/480 VAC. Rated frequency is 50/60 Hz. Power rating is 50 kVA continuous, 60 kVA instantaneous. The PDU mounts in a standard 19-inch rack.



ACGD Power Distribution Unit
Illustration 1-1

1-2 Scope

This manual describes procedures for installation and basic operation of the HFD Power Distribution Unit as well as procedures for failure diagnosis/isolation and installation of field replaceable components of the PDU.

2- SPECIFICATIONS

Specifications subject to revision without notice

GE Part Number
5113474

2-1 Electrical Specifications

Power Rating

50 kVA Continuous; 60 kVA Instantaneous

Input Voltage

200/208/380/400/415/480 VAC Delta, User Selectable

Output Voltage

208/120 VAC WYE and 420 VAC Delta

Overload Protection

Input main circuit breaker with adjustable trip settings; output distribution circuit breakers

Load Current

69 Amps total maximum continuous (208V)

34 Amps total maximum continuous (420V)

Transformer Impedance

Less than 2%

Efficiency

98% Typical

2-2 Environmental Specifications

Heat Dissipation

1,995 BTU/Hr

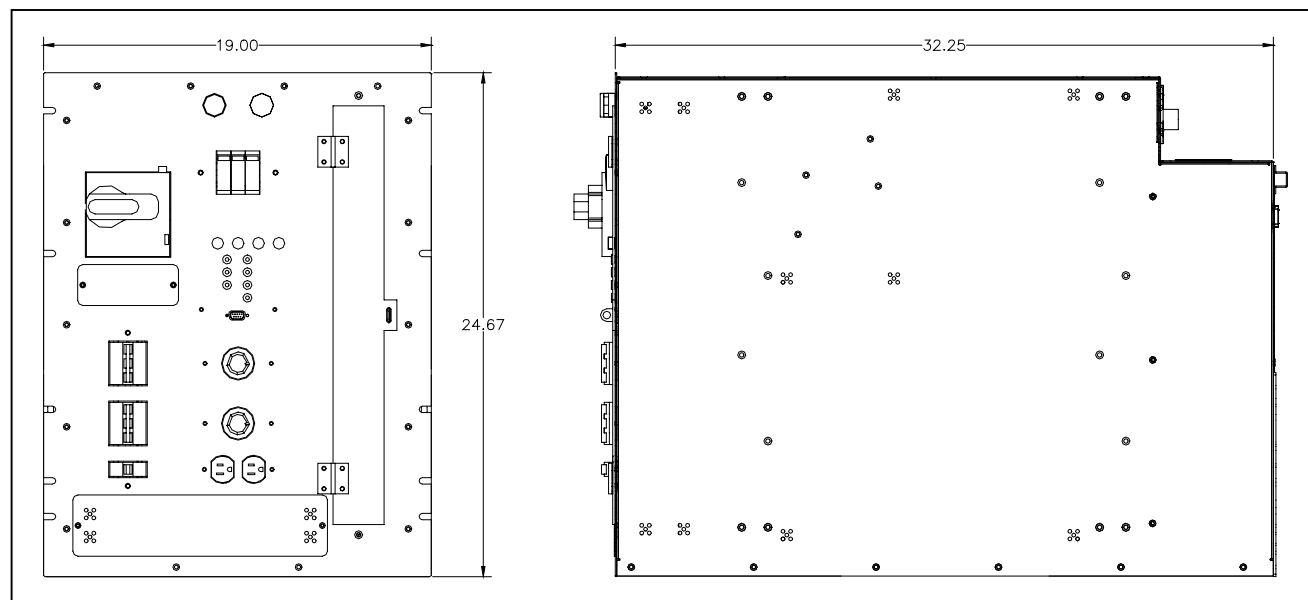
2-3 Mechanical Specifications

Weight

680 lbs. / 308 kg.

Dimensions

See Illustration 2-1

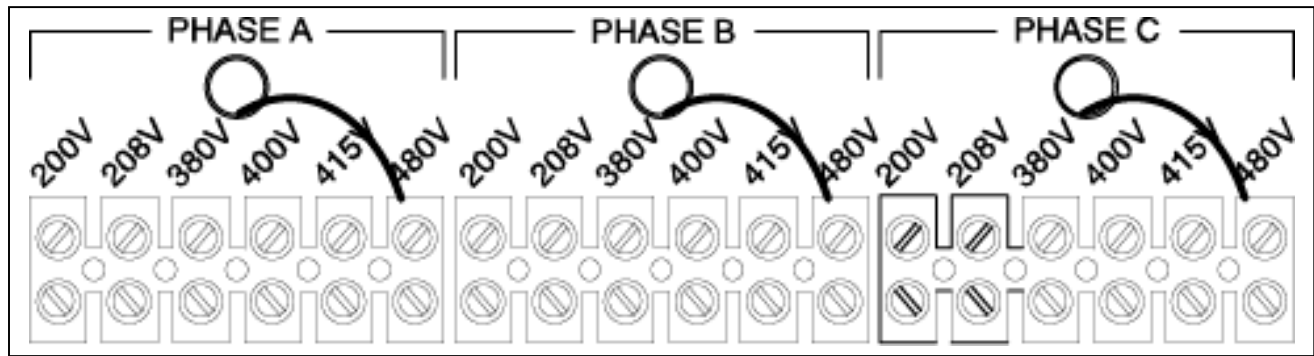


HFD PDU Overall Dimensions
Illustration 2-1

3- INSTALLATION AND OPERATION**3-1 Installation****3-1.1 Input Voltage Selection**

Before connecting the input power cables, determine the nominal value of the input voltage. The PDU allows for nominal input voltages of 200, 208, 380, 400, 415, or 480 VAC (three phase). Input voltage selection is accomplished as follows:

1. On the Main Disconnect Panel (MDP), turn off the PDU circuit breaker, lock and tag appropriately.
2. Using a voltmeter or other voltage indicating device at the Main Input Terminal Board, check to be sure that no voltage is being applied to the input of the PDU.
3. Remove the cover panels from the front panel of the PDU marked "MAIN INPUT CIRCUIT BREAKER CURRENT ADJUST" and "INPUT VOLTAGE SELECT".
4. Illustration 3-1 shows the Input Voltage Selection terminal block. For each of the three phases, be sure the wire is connected to the proper terminal for the actual input voltage at the site and that the terminal screw is tight to make a good connection.



INPUT VOLTAGE SELECT TERMINAL BLOCK

Illustration 3-1

5. The overload and short circuit trip settings for the Input Circuit Breaker must be set to the correct values corresponding to the input voltage.
 - 5a. Remove the small cover plate under the main circuit breaker which is held in place by two screws and lift open the transparent plastic cover on the lower portion of the circuit breaker to give access to the dip switches controlling the circuit breaker operation.
 - 5b. Illustration 3-2 shows the position of the dip switches for each input voltage selection. The “L” and “I” switches must match the configuration for the incoming voltage used as shown in illustration 3-2.

3-1.2 Connection

Note the Cable Access Panel shown in Illustration 3-5 at the lower portion of the rear of the PDU. This panel provides strain relief cable clamps for the input and the various output cables. This panel may be removed from the PDU to provide easier access to the terminal blocks.

1. Route the cables through their respective cable clamps and connect cables to the appropriate terminals as shown.
2. Replace the Cable Access Panel and tighten the cable clamps.

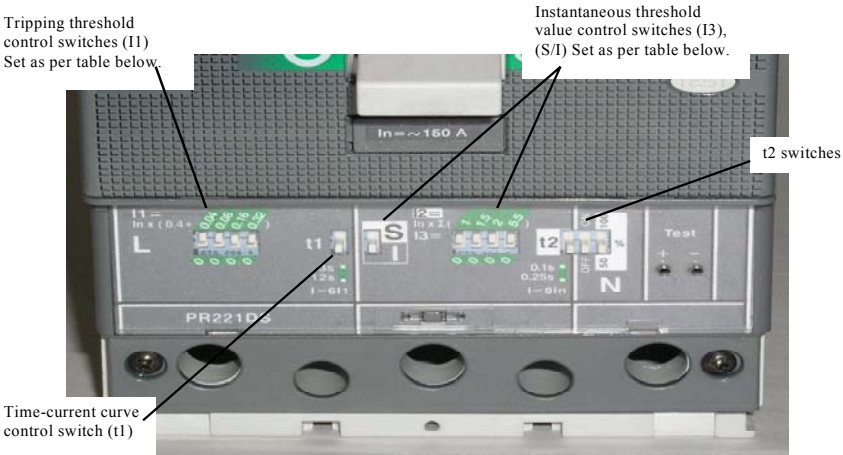
⚠ IMPORTANT

For input cables, use wire sizes in accordance with National Electric Code Table 310-16. The following table gives maximum full-load input currents in amperes (which would occur at 10% undervoltage) for each of the input voltages, as a guide to selecting proper wire sizes.

Maximum Input Current vs. Input Voltage

Table 3-1

Input Voltage	200V	208V	380V	400V	415V	480V
Maximum Input Current	143A	137A	75A	71A	69A	60A



INPUT VOLTAGE	200	208	380	400	415	480
"L" SWITCH SETTINGS						
"I" SWITCH SETTINGS						

CIRCUIT BREAKER DIP SWITCH SETTINGS
Illustration 3-2

3-2 Operation

3-2.1 Main Input Circuit Breaker

The Main Input Circuit Breaker operating handle has three positions:

- OFF (as shown in Illustration 3-4),
- ON (vertical), and
- TRIPPED (slightly to the right of vertical). If the Main Input Circuit Breaker has been tripped, it is necessary to move the handle to the OFF position to reset it before turning it ON.

The Main Input Circuit Breaker may be locked in the OFF position. By pressing on the small triangle on the operating handle, the lockout tab is made accessible to a padlock.

3-2.2 Power Off

This pushbutton at the top of the front panel, when pressed, will immediately trip the Main Input Circuit Breaker, interrupting all power to the load.

3-2.3 EMO Reset

This pushbutton immediately to the right of the Power Off pushbutton will enable the Emergency Stop circuit.

3-2.4 TVSS Indicators

To the right of the Main Input Circuit Breaker is a small cover plate labeled “TVSS INDICATORS”. A cutout in this plate makes visible the indicator on each of the three TVSS modules. Should a TVSS module fail because of a particularly large voltage surge, a red flag on the module indicator will appear and the Main Input Circuit Breaker will trip. Therefore, should the Main Circuit Breaker trip unexpectedly, check the TVSS Indicators for one or more failed modules showing the red flag. Any module that fails must be replaced to restore the PDU to operation with surge protection functioning. See Section 4-1.1, Replacement of a TVSS Module, for instructions for replacing a TVSS Module.

3-2.5 Phase Indicators

Just below the TVSS Indicator cover plate are three Phase Indicator lights. These indicate that the transformer is energized and ready to deliver power to the load.

3-2.6 Phase Reversal Relay

The PDU has a phase reversal function to ensure correct phasing is present for proper fan rotation. If incorrect phasing is applied to the unit, the e-stop circuit will not activate.

3-2.7 J5, J7, & J8 Pass-thru Connector

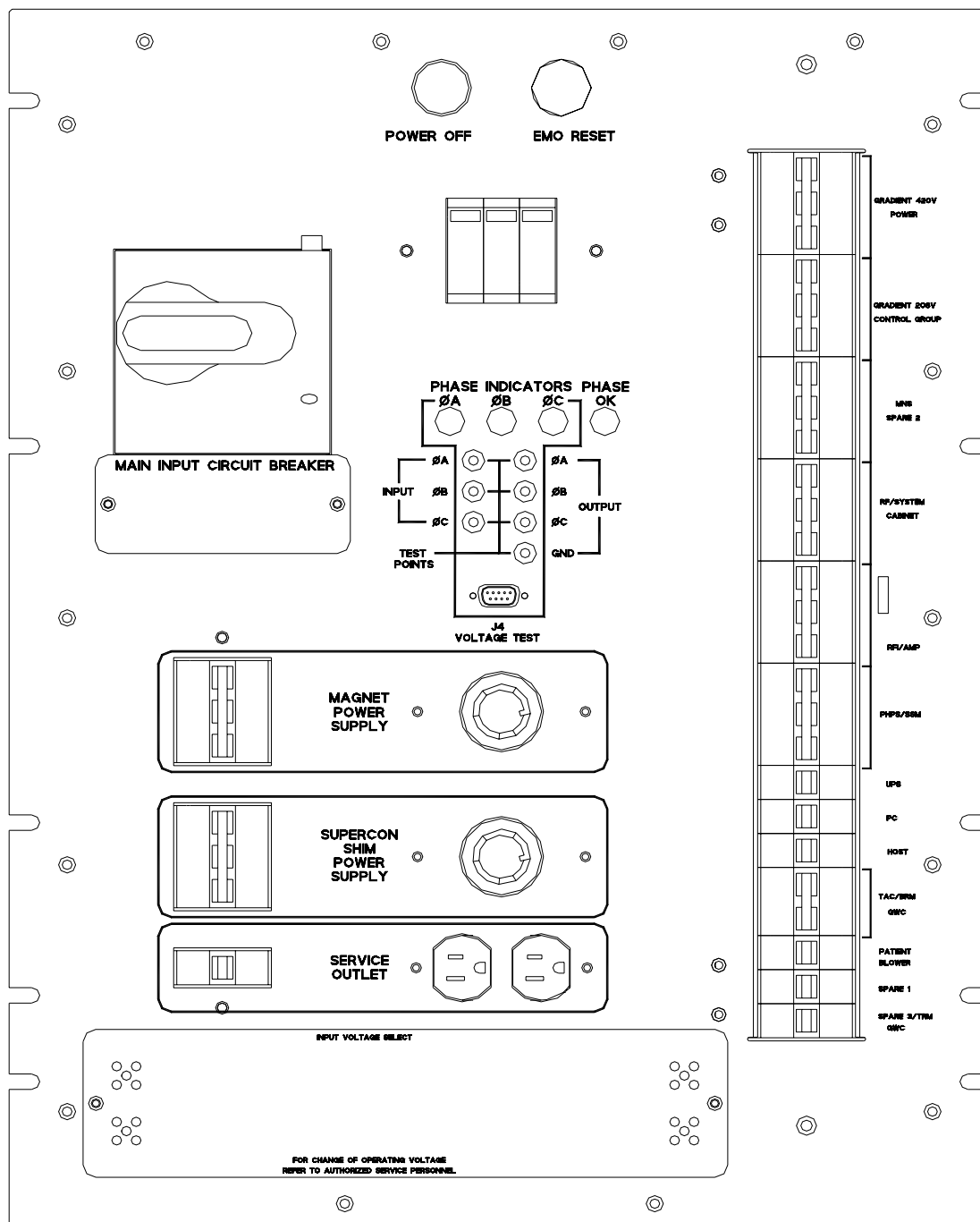
The PDU has three (3) pass-thru, 9 pin, Sub D connectors to allow a cable to be run inside the cabinet from the GP to the back panel of the PDU.

3-2.8 J6 Pass-thru Connector

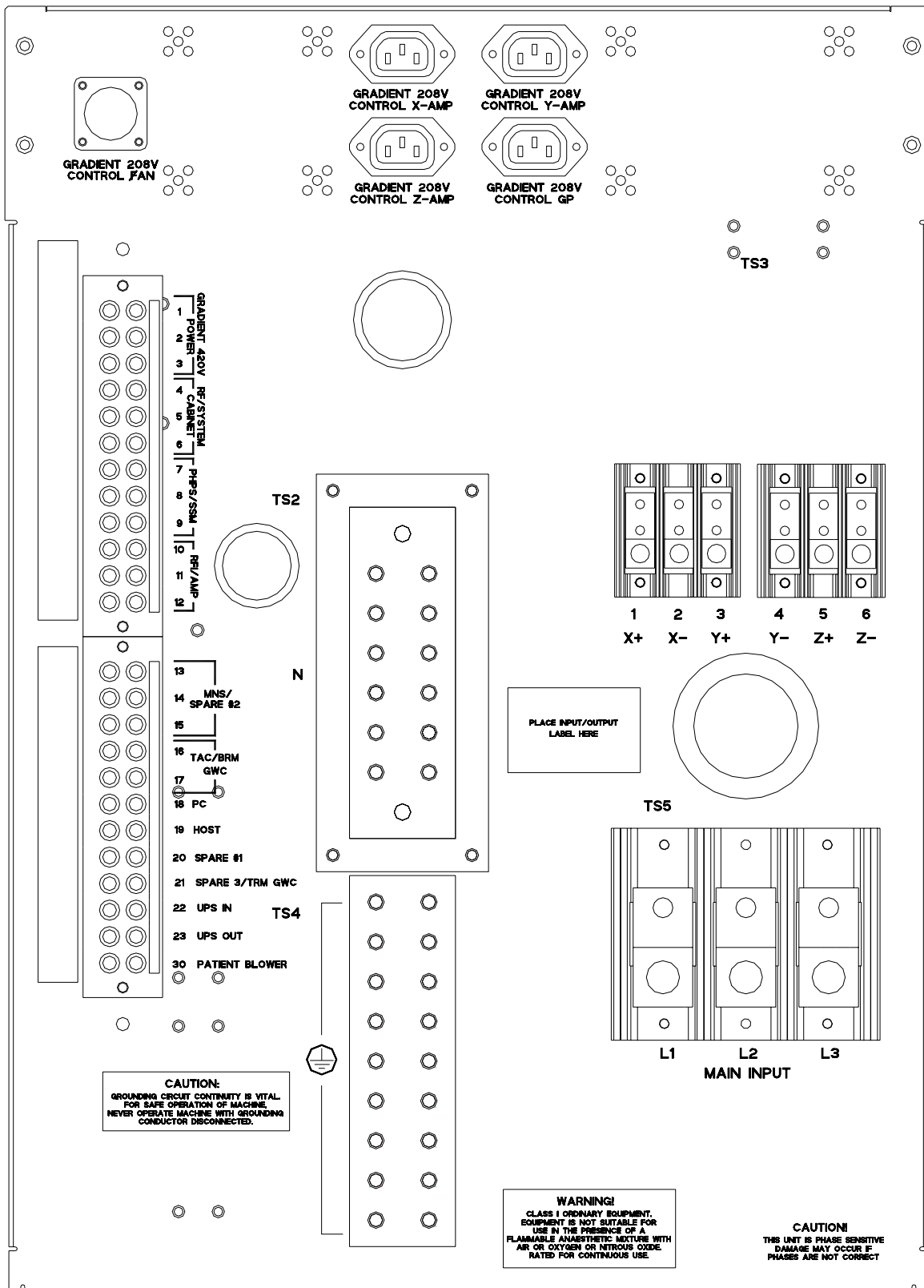
The PDU has one (1) pass-thru, 15 pin, Sub D connector to allow a cable to be run inside the cabinet from the GP to the back panel of the PDU.

3-2.9 Input / Output Voltage Test Points

Immediately below the Phase Indicators on the front panel are jacks into which can be plugged connectors for testing for input and output voltage. See Illustration 3-4. The voltage at these points are reduced by a factor of 100:1 from the actual voltage at the terminal blocks.



FRONT PANEL COMPONENT LOCATIONS
Illustration 3-4



REAR PANEL COMPONENT LOCATIONS
Illustration 3-5

4- FIELD REPLACEABLE UNITS (FRU'S)

LISTING OF FRU's

Table 4-1

Item	Description	GEMS No.	Teal Part No.	FRU Type
1	TVSS Module	2301200-3	4990387	1
2	Control PCB	2374930-2	4990604	1

4-1 FRU Installation

4-1.1 Replacement of the TVSS Module

The operation of the TVSS Indicators is explained in Section 3-2.4, TVSS Indicators. If one (or more) of the indicators shows that a TVSS Module has failed, follow the steps below to replace the failed module and restore TVSS protection:

1. On the MDP, turn off the PDU circuit breaker, lock and tag.
2. Using a voltmeter or other voltage indicating device, check the voltage at the Main Input Terminal Board to be sure that no voltage is being applied to the input of the PDU.
3. On the front panel of the PDU, remove the small cover plate over the TVSS Modules.



TVSS Cover Plate

TVSS COVER PLATE

Illustration 4-1

4. Then pull out the failed TVSS Module(s) and plug in the replacement(s) as shown in Illustration 4-2.



TVSS MODULE REPLACEMENT

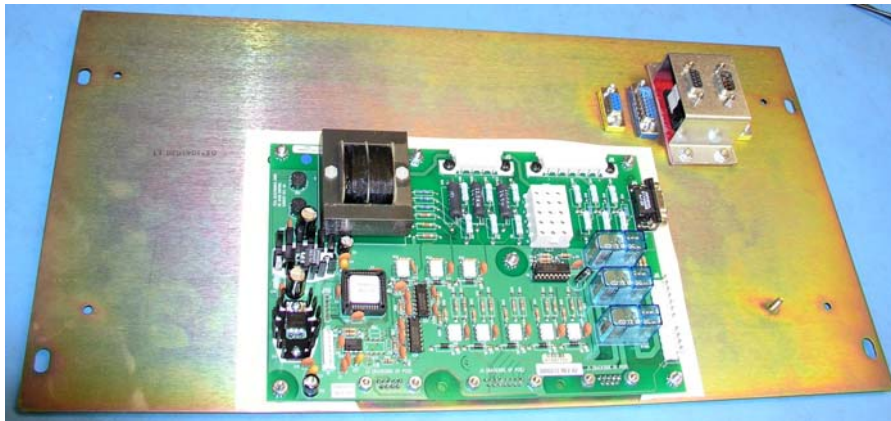
Illustration 4-2

5. Replace the TVSS Cover Plate and re-energize the PDU.
6. Verify that all TVSS Modules are good (no red flags displayed).

4-1.2 Replacement of the Control PC Board

If the control board is deemed defective as per the Troubleshooting Guide in 4-1.4 follow the steps below to replace the control board. The control board is mounted on the inside of the back panel.

1. Turn off the main input circuit breaker.
2. Remove the rear connector panel.
3. Pull assembly away from back panel enough to remove connectors to the control board.
4. Remove control board from it's mounting locations.
5. Install new control board at the nylon standoff mounting locations.
6. Connect the connectors removed in step 3. back onto the control board.
7. Install rear connector panel.
8. Turn on main input circuit breaker.



Control PC Board, Internal View
Illustration 4-3

4-1.3 Replacement of the SGA PC Board

If the SGA board is deemed defective as per the Troubleshooting Guide in 4-1.4 follow the steps below to replace the SGA board. The SGA board is mounted to the SGA terminal block located on the inside, rear panel.

1. Turn off the main input circuit breaker.
2. Remove the rear connector panel.
3. Remove the clear lexan safety cover.
4. Remove control board from it's mounting locations.
5. Install new control board at the mounting locations.
6. Install the clear lexan safety cover.
7. Install rear connector panel.
8. Turn on main input circuit breaker.



Internal View
Illustration 4-4

4-1.4 Troubleshooting Guide

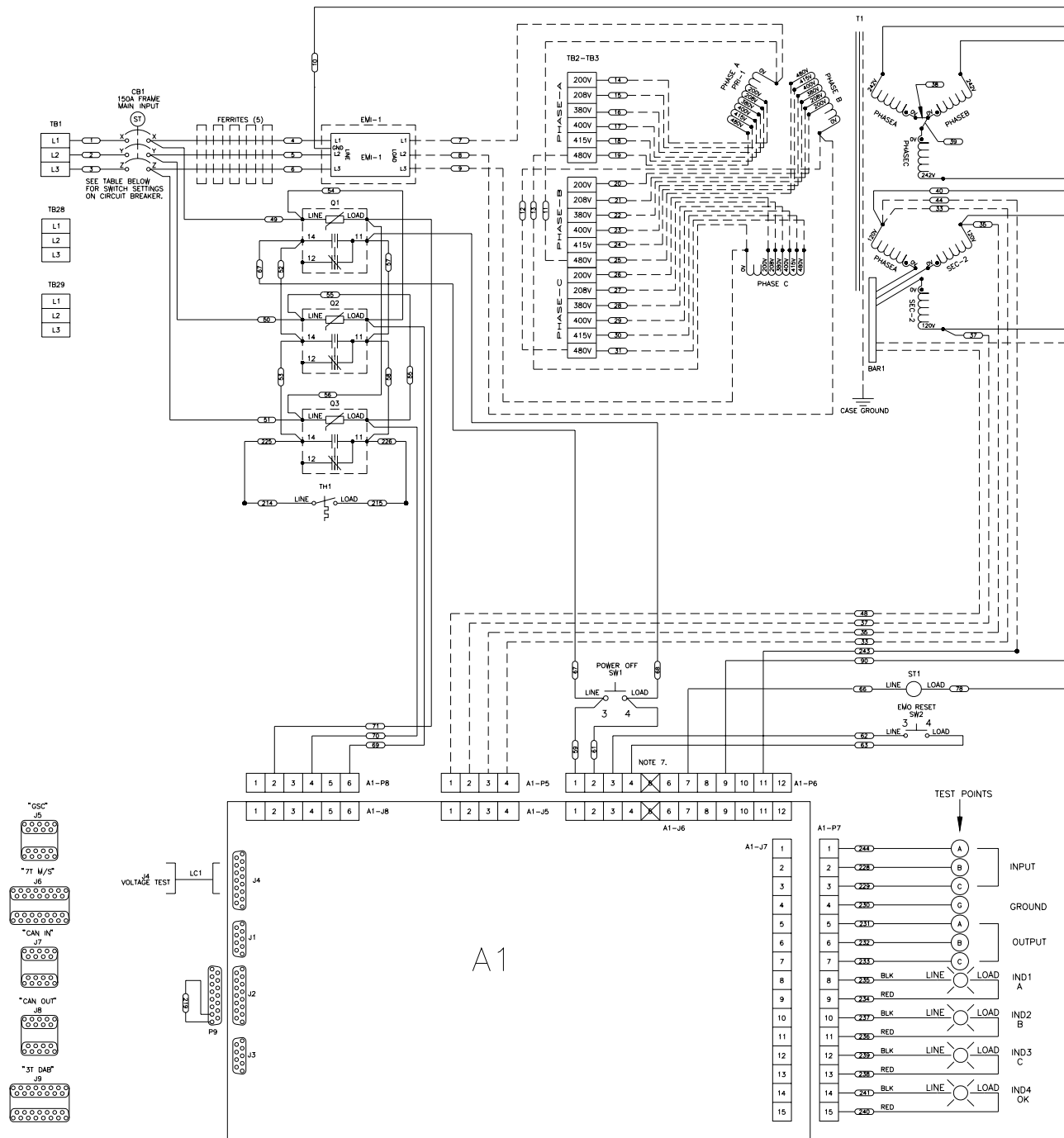
The following table lists fault conditions that may be encountered along with possible causes and solutions.

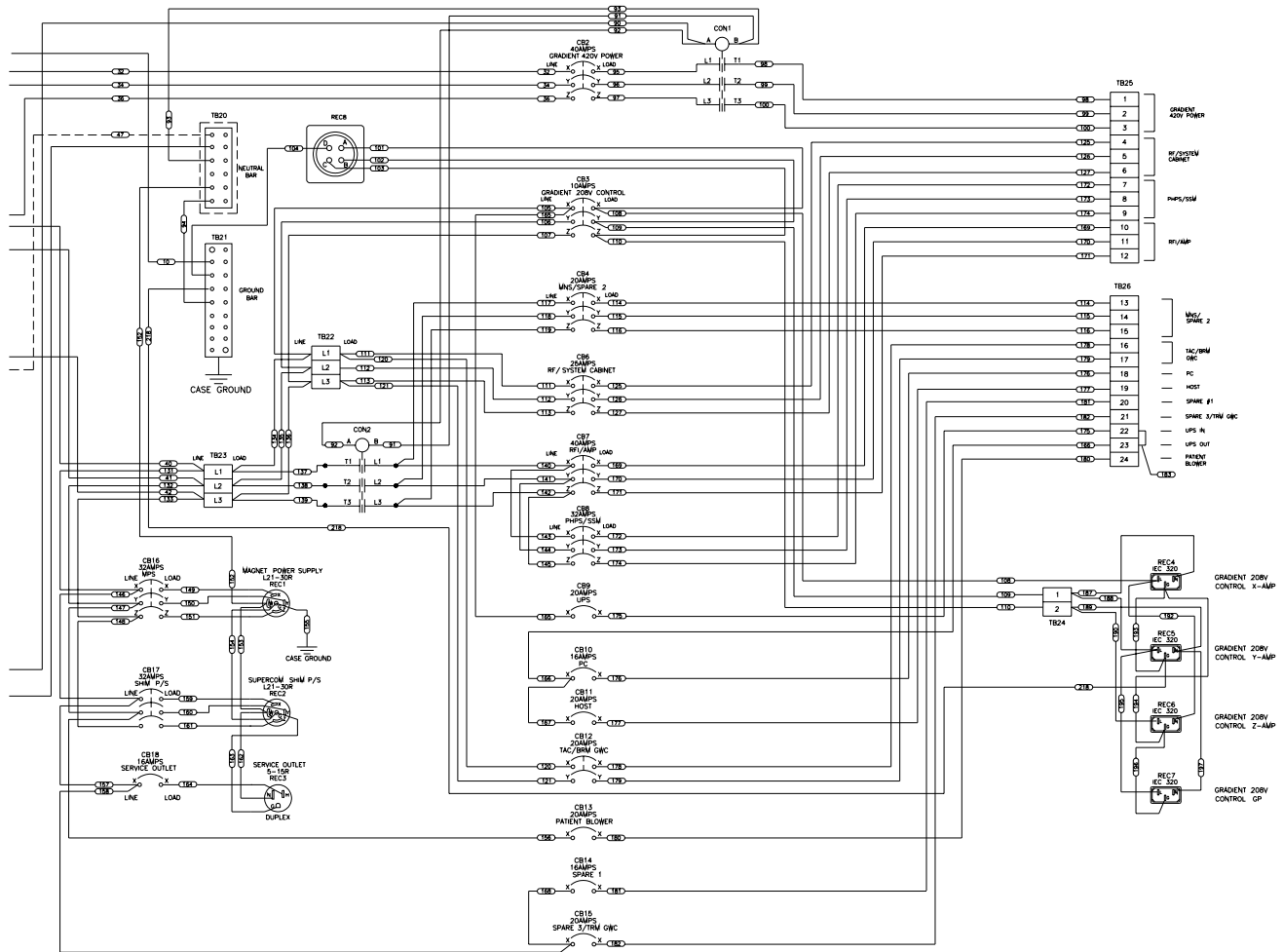
Troubleshoot Guide
Table 4-2

MODULE	SYMPTOM	CAUSE	SOLUTION
PDU	Main Input CB Trips	TVSS Module Failure	1. Check TVSS Modules for failure. A red "flag" will appear in TVSS module window when failure has occurred. 2. Replace defective TVSS module.
		Overtemperature	1. Investigate cause of overtemperature 2. Allow unit to cool down. 3. Reset the circuit breaker
		Shorted Power Off Switch	1. Turn power off 2. Reset Main Circuit Breaker 3. Turn power on 4. If CB trips, replace PDU
		Open from J3-2 to J3-3	1. Disconnect cable at J3 2. With DVM check for open at cable connector P3-2 to P3-3.
	Distribution Breakers Tripping	Defective Circuit Breaker	1. Turn power off 2. Check if circuit breaker still trips (won't latch up)
		Overload	1. Check load by disconnecting cabinet from PDU whose circuit breaker is tripping, or use an Amp Clamp / DVM to check current draw. 2. Correct load problem.
	CON1/CON2 Contactor not operating	Defective Contactor	1. Ensure that the contactor CON1 is receiving 120V to the coils. 2. If 120V is present, replace the contactor. 3. If 120V is not present, verify that CON2 is receiving 120V to the coils 4. If 120V is present, replace the conatctor. 5. If 120V is not present, replace the PDU.

		Defective Control Board A1	1. Disconnect cable at J2. 2. Check at cable connector if P2-7 to P2-8 is shorted. 3. If shorted, replace PDU 4. If not shorted, connect test plug and verify PDU operation. 5. If PDU operates with Test Plug, then replace PDU. 6. If PDU does not operate with Test Plug, then replace Control Board A1
		Incorrect Phasing	1. Turn power off to PDU 2. Re-connect input wires with proper phasing
	Indicators not lighting when power is applied	Input Circuit Breaker off	1. Turn Input Circuit Breaker, CB1, on.
		Input Phase Missing	1. Check that other phase LED's are on. 2. Replace PDU
		Light Burned Out	1. Check that other phase LED's are on. 2. Replace PDU
		Not Getting Proper Output	1. Measure input and output, at front panel test points, of each phase, from phase to phase (208 VAC: 208 VAC) and phase to ground (120 VAC: 120 VAC). 2. If difference is not a 100:1 ratio, replace PDU

SCHEMATICS





REVISION HISTORY

REV	DATE	AUTHOR	PRIMARY REASONS FOR CHANGE
A0	10/18/04	T. Cole	New Release.
A1	1/07/05	T. Cole	Corrected error in Section 3-1.1, Step 5b on setting input breaker switches.